Problem Chosen	2026 MCM/ICM Summary Sheet	Team Control Number 2609080
D		

Beyond the Scoreboard: An Integrated Dynamic Optimization Framework for Franchise Evolution

Summary

Amidst the explosive commercial growth of the WNBA, franchises must balance competitive success with financial sustainability. We develop an integrated strategic framework for the Las Vegas Aces to optimize multi-stage decision-making and maximize total returns.

In Task 1, we develop the **Dynamic Optimization Model (DOM-LV)** to decouple the structural contributions of competitive performance and commercial operations to total returns. Through **Lagrangian-based constrained optimization** and **sensitivity analysis**, we identify that the **marginal return** on commercial investment significantly outweighs that of player salaries. Benchmark simulations reveal that under parameters α=0.2 and β=0.5, the system achieves an **optimal total utility of $78.57M**, ensuring data-driven resource optimization under regulations.

In Task 2, we develop the **Scoring-Based Value Projection Model (SB-VPM)** for roster optimization. To address data scarcity, **Monte Carlo Simulation** is utilized to expand the candidate pool, ensuring statistical robustness. By constructing a "Net Value" index, we identify high-impact players and confirm that on-court performance and key accolades outweigh cost factors as primary value predictors.

In Task 3, we employ **scenario-based simulation** to characterize the impact of league expansion by adjusting key mechanisms such as salary inflation, playoff target, flow coefficient, initial salary, and high distance assumption. The results quantify the strategic efficacy of diverse developmental pathways under expansion: scores for the top five decision schemes range from 10.27 to 10.62, while the bottom five cluster between 7.63 and 8.17. This data-driven ranking identifies the most resilient developmental strategies for navigating competitive and revenue structural shifts.

In Task 4, we propose a **Zonal Dynamic Pricing Model** integrating **Fan Lifetime Value (LTV)**. By optimizing price-volume trade-offs across eight market scenarios, the model achieves an **average utility gain of 62.30%**, with single-game revenue peaking at **$6.88M** under the "Clark Effect." This framework provides an actionable strategy for maximizing immediate returns while ensuring sustainable franchise growth.

Finally, we integrate the four modules into a unified decision-support framework and present the recommended strategy in a letter to the team’s owner.

Keywords: Dynamic Optimization; Lagrangian Method; Monte Carlo Simulation; Strategic Resilience; Zonal Dynamic Pricing

Contents

1 Introduction 3

1.1 Problem Background 3
1.2 Restatement of the problem 4
1.3 Our Work 4

2 Assumptions and Justifications 4

3 Task 1: Dynamic Optimization of Franchise Value and Profitability 5

3.1 Notations 5
3.2 Dynamic Optimization of Franchise Value and Profitability 6
3.3 The Solution of Model 1 9

4 Task 2: Player Value Evaluation and Roster Optimization Model 11

4.1 Notations 11
4.2 Player Value Evaluation and Roster Optimization Model 11
4.3 The Solution of Model 2 15

5 Task 3 : Player Net Value Evaluation under League Expansion 15

5.1 Model Overview & Objectives 15
5.2 Notations 16
5.3 Stochastic Player Performance Generation 16
5.4 Mathematical Formulation of Market Effects 17
5.5 The Solution of Model 3 18

6 Task 4: Zone-Based Dynamic Pricing Model with LTV Integration 19

6.1 The “Profit-Acquisition” Trade-off 19
6.2 Notations 19
6.3 Model construction 20
6.4 The Solution of Model 4 20

7 Model Evaluation and Further Discussion 21

7.1 Strengths 22
7.2 Weaknesses 22
7.3 Further Discussion 22

8 Letter 23

9 References 25

1 Introduction

1.1 Problem Background

Women's professional sports are rapidly integrating into the global commercial mainstream, fueled by enhanced visibility and significant investment. Currently, the WNBA is at a historic crossroads. From 2024 to 2025, the league's total valuation tripled, with average team values jumping 180% to $269 million. Notably, the WNBA's valuation-to-revenue multiple now stands at 12x, exceeding that of the NWSL (6.8x) and even the NBA (11.9x). Such rapid financial expansion defines a new era for the league, where immense growth potential coexists with emerging structural challenges.

The Las Vegas Aces (Aces) serve as the vanguard of this transformation by exemplifying a synergy between competitive excellence and commercial success. Between June 2024 and June 2025, the team's valuation surged from $140 million to $310 million, representing a 121% year-over-year increase. While annual revenue grew from $17.8 million to $22 million, the franchise maintained its position at the forefront of league valuations despite the intricate trade-offs between valuation multiples and athletic investment.

In terms of market performance, the Aces made history during the 2024 season as the first WNBA team to sell out all regular-season home games. Although their home venue has a fixed capacity of 8,600, the team achieved an average attendance exceeding 9,500 per game through strategic venue scheduling. This 18% increase in attendance underscores a high correlation between on-court dominance and commercial viability, establishing the Aces as an ideal empirical case for studying dynamic optimization models in professional sports.

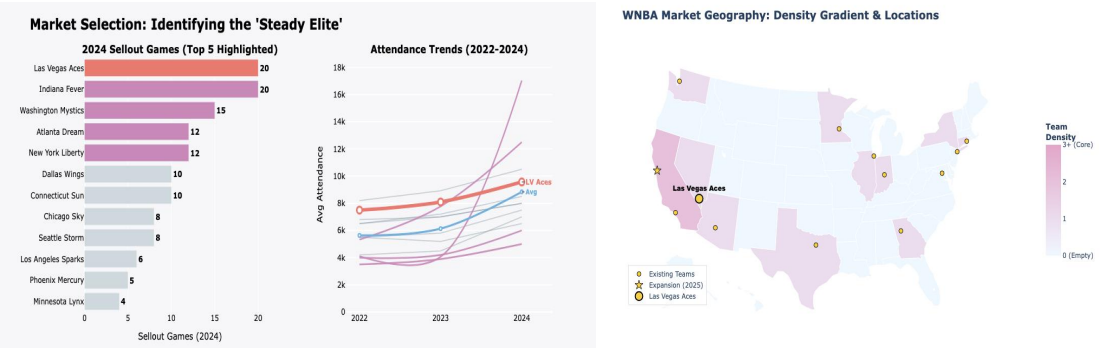


Figure 1: WNBA Attendance Trends and Franchise Geographical Distribution

However, elite on-court performance now comes with increasingly complex managerial challenges. Under the current Collective Bargaining Agreement, the 2025 salary cap increased by only 3% to $1,507,100, while the players' union's early opt-out introduces substantial uncertainty for post-2026 salary structures. This volatility limits the effectiveness of traditional financial planning. At the same time, the 2026 expansion introducing the Toronto Tempo and Portland franchises allows teams to protect only six players, exposing the Aces to talent dilution and potential losses within their core rotation.

For Aces management, the central challenge is to systematically evaluate and quantify the full value of star players, encompassing both on-court impact and brand contribution, under heightened budget uncertainty. In the short term, these decisions will shape competitive stability; in the long term, they represent a reinvestment in brand equity and financial sustainability that will ultimately determine the durability of the franchise's competitive advantage.

1.2 Restatement of the problem:

Considering the current situation in Las Vegas Aces and the information outlined in the problem statement, we need to solve the following problems:

Problem 1: We develop a multi-objective dynamic decision-making model tailored to the Aces' operational environment, incorporating key drivers of franchise development, including win-rate–driven game revenue, commercial revenue linked to multi-season performance and fan base size, and attendance adjusted by price elasticity.

Problem 2: To mitigate expansion-related shocks, we construct a dynamic evaluation framework that quantifies talent dilution, salary inflation, and market diversion to support roster optimization. By integrating region-based Fan Lifetime Value (LTV) pricing, the model balances short-term revenue generation with long-term fan acquisition costs under structural uncertainty.

Problem 3: Leveraging the model results, we present a strategic letter to the Aces' executive team that translates empirical insights into actionable recommendations, explicitly addressing trade-offs among competitive performance, financial returns, and risk management.

1.3 Our Work:

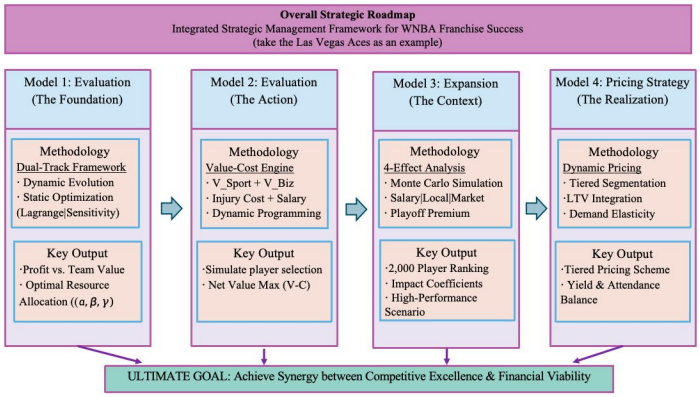


Figure 2: Our Overall Work

2 Assumptions and Justifications

Assumption 1: An increase in a team's win rate leads to growth in game-related revenue, with a nonlinear amplification effect. Specifically, marginal improvements in win rate generate disproportionately larger revenue gains at higher win-rate levels.

Justification: This assumption reflects the observed reality in professional sports, where stronger teams command higher commercial premiums during late-season and high-stakes games. The nonlinear relationship between win rate and revenue is therefore modeled using a second-order polynomial to capture the marginal amplification effect.

Assumption 2: The team generates fixed revenues and incurs fixed operating costs during the season that are not directly tied to individual game outcomes. Fixed revenues primarily arise from league-wide media broadcasting agreements, while fixed costs include venue leasing, administrative expenses, and logistical operations.

Justification: These revenue and cost components are typically determined by league agreements and long-term contracts prior to the season and therefore do not fluctuate with game frequency or win–loss results in the short run. As such, they serve as structural financial constraints within the model.

Assumption 3: The evolution of the fan base is jointly driven by on-court performance and star power, and exhibits dynamic accumulation across multiple periods.

Justification: This assumption captures brand stickiness in sports consumption markets. Win rate sustains engagement among the existing fan base, while player popularity acts as a growth driver, together shaping future monetization potential and long-term franchise valuation.

Assumption 4: Injuries are modeled as stochastic loss events with a specified probability distribution, and their impacts on team performance are asymmetric.

Justification: The model incorporates an injury risk term that converts unexpected injuries into expected performance losses, and employs Monte Carlo simulations to differentiate impact weights between star players and non-star players. This structure realistically captures the leftward shift and increased volatility in team performance distributions that arise when core assets are impaired.

Assumption 5: Ticket sales are governed by an adjusted log-linear demand function, in which a market interest parameter effectively captures extreme fluctuations in the external environment.

Justification: By introducing external coefficients, this assumption translates market shocks, such as the “Clark Effect,” into quantifiable demand-shift factors. The log-linear specification accurately reflects price elasticity under high-demand conditions while accounting for venue capacity constraints.

3 Task 1: Dynamic Optimization of Franchise Value and Profitability

3.1 Notations

Symbol	Description
δ	Discount factor, used to calculate the present value of future earnings.
Π_t	Net profit of the team in season t.

$V_{franchise,t}$	Total market franchise valuation of the team at time t.
R_{game}	On-court driven revenue, directly determined by win rate and game performance.
$x_{i,t}$	1 or 0, depend on whether the player in this team
R_{biz}	Business-driven revenue, referring to non-competitive income generated by star appeal.
P_i	Individual commercial influence index of player i.
W_t	Winning percentage in season t.
F_t	Total fan base at the beginning of the season, representing the team's brand equity stock.
η	Price elasticity of demand for tickets, used to balance attendance and revenue.

Table 2: Notations used in 4.2

3.2 Dynamic Optimization of Franchise Value and Profitability

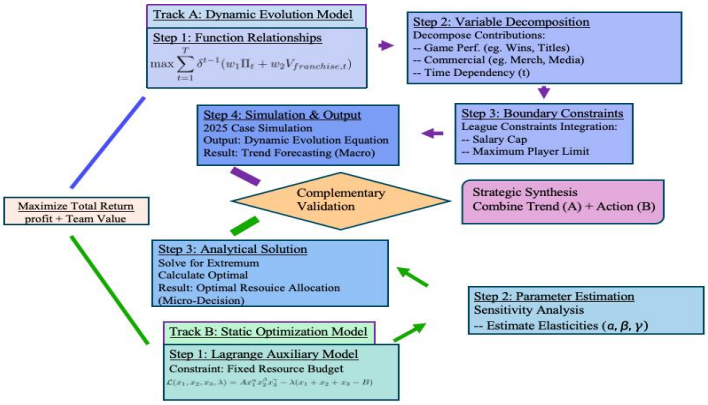


Figure 3: Framework of the Dual-Track Evaluation Model

We constructed a multi-objective dynamic optimization model. This model focuses not only on annual net profit Π_t but also incorporates long-term franchise valuation $V_{franchise}$ into the objective function, reflecting the high-premium characteristics of the current WNBA capital market.

3.2.1 The Global Objective Function

The overarching objective function we established aims to capture the dual attributes inherent in professional sports management: the profitability of the team as a business entity and the appreciation of the team as a capital asset.

We abstract the management's decision-making goal as the maximization of the weighted Net Present Value over the planning horizon:

$$\max \sum_{t=1}^{T} \delta^{t-1}(w_1 \Pi_t + w_2 V_{franchise,t})$$

where δ is the discount factor used to account for the time value of money and potential operational risks; ω_1 and ω_2 represent the weighting coefficients that reflect management's relative emphasis on immediate profit versus future brand equity.

3.2.2 Dynamic Optimization of Short-term Economic Benefits

At the short-term operational level, the core of the model is to maximize the net profit Π_t for each season. Unlike traditional static revenue models, drawing on the financial characteristics of Aces, we develop a refined modeling framework for the profit structure as follows:

$$\Pi_t = R_{fixed} + R_{game}(W_t) + R_{biz}(W_t, x_{i,t}) - \sum_{i=1}^{N} C_i x_{i,t} - C_{fixed}$$

This formula quantifies the team's cash flow performance through four core dimensions:

Fixed Revenue R_{fixed} represents guaranteed income determined before the season begins, with the core component being the league-wide shared media rights agreement revenue. Considering the new round of media rights contracts expected to take effect after 2026, which are projected to grow four to five times from current levels, this provides the most stable financial "bottom pool" for the team to cope with salary inflation. It also serves as a critical buffer variable in the model against policy fluctuation risks.

On-court Driven Revenue $R_{game}(W_t)$ mainly indicates that the higher the winning rate, the more exponentially increased returns it can bring

$$R_{game}(W_t) = \alpha_0 + \alpha_1 W_t + \alpha_2 e^{W_t}$$

Star Commercial Value We define this as a function driven by the dual factors of win rate and individual star appeal:

$$R_{biz}(W_t, x_{i,t}) = \beta_0 + \beta_1 W_t + \beta_2 \sum_{i=1}^{N} P_i x_{i,t}$$

Expenditure and Constraints: The model calculates final profit by subtracting player salary expenditures $\sum C_i x_{i,t}$ and fixed operating costs C_{fixed}. This variable must be optimized under multiple hard constraints. It is strictly limited by the league's salary cap C_{cap}: $\sum_{i=1}^{N} C_i x_{i,t} \leq C_{cap}$. Additionally, the roster size must comply with league administrative regulations: $N_{min} \leq \sum x_{i,t} \leq N_{max}$.

3.2.3 Evolution of Long-term Franchise Valuation

Considering the high growth potential of WNBA team valuation multiples, the long-term value $V_{franchise}$ is a key indicator of operational success. We constructed a valuation evolution model based on star effects and the fan economy:

$$V_{franchise,t} = \theta_1 \sum_{i=1}^{N} P_i x_{i,t} + \theta_2 \bar{W}_t + \theta_3 \log(F_t)$$

$\bar{W}_t$ can be viewed as the average winning percentage of the past.

3.2.4 Cost Dimension Refinement

To further refine the strategic trade-offs mentioned in Model 4.2.1, we must determine the optimal distribution of the total budget B across different functional departments. Following the economic modeling logic, we define the team's output utility as a Cobb-Douglas utility function concerning inputs from three core departments:

$$U(x_1, x_2, x_3) = A \cdot x_1^{\alpha} \cdot x_2^{\beta} \cdot x_3^{\gamma}$$

x_1 represents players and salary investment, x_2 represents branding and marketing

Investment and x_3 represents venue and experience investment

Under the total budget constraint B, we construct the Lagrangian function to solve for the extreme point of utility maximization:

$$\mathcal{L}(x_1, x_2, x_3, \lambda) = A x_1^{\alpha} x_2^{\beta} x_3^{\gamma} - \lambda(x_1 + x_2 + x_3 - B)$$

Taking player investment as an example, the optimal solution is: $x^* = \frac{\alpha}{\alpha+\beta+\gamma} B$

3.3 The Solution of Model 1

3.3.1 Base-Year Simulation and Dynamic Trade-off Analysis

The core of this model is a dynamic trade-off framework designed to synchronize the maximization of long-term team value and short-term profitability by optimizing the team's revenue. To capture the time-evolving characteristics of this process, we utilized the actual financial statements and operational data from the 2025 season as a simulation baseline and through numerical solutions of the 2025 baseline scenario, we obtained the following waterfall chart, which illustrates the detailed financial flow:

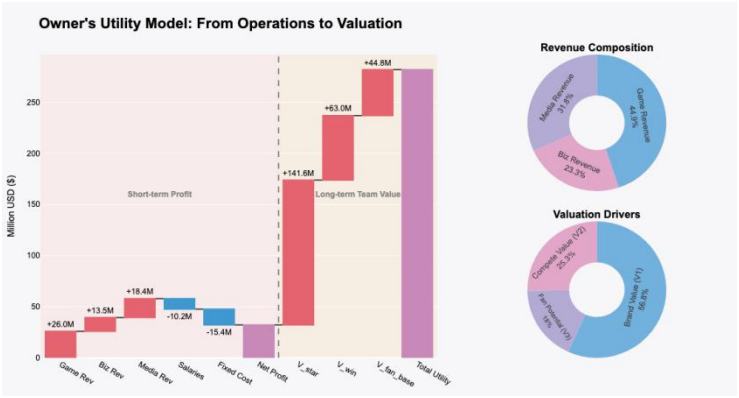


Figure 4: Owner's utility model:From Operations to Valuation

This chart can make people intuitively feel that short-term income is mainly dominated by competition income, media rights and business development. The annual net profit growth is restricted due to the constraints of salary and fixed costs. In contrast, the long-term valuation increased significantly by 249.4 million USD.

3.3.2 Lagrange Auxiliary Model for Resource Optimization

Since the dynamic model primarily focuses on trend prediction, we introduced the Lagrange Auxiliary Model to perform static optimization seeking for concrete resource investment guidance. In this model, because the specific values of parameters α, β, γ are initially unknown, we first estimated them through sensitivity analysis.

3.3.2.1 Parameter Sensitivity Estimation

Considering that fixed expenditures such as venue rent, electricity, and infrastructure maintenance corresponding to γ possess strong rigidity in the short term and are difficult to change quickly, we fixed the weight γ at 0.3. Subsequently, we plotted the functional

relationship between the weights of player investment α and marketing investment β regarding their impact on total revenue.

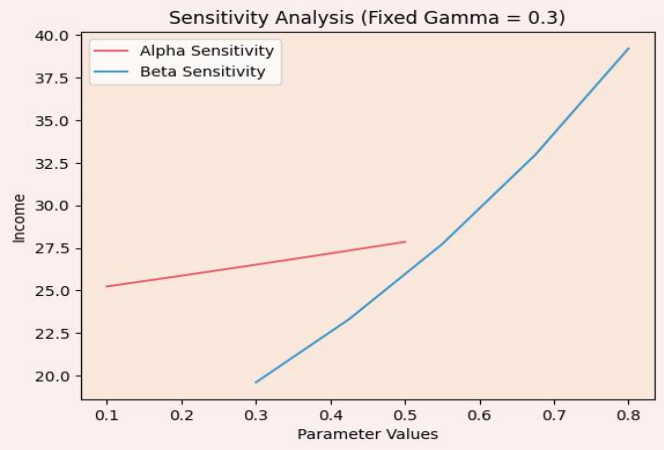


Figure 5: Sensitivity Analysis for Lagrange Auxiliary Model

It is observable from the graph that the impact slope of the marketing weight β on total revenue is significantly greater than that of the player weight α.

3.3.2.2 Optimal Capital Allocation Ratios

and based on the expenditure simulation for 2024, we solved for capital allocation under various scenarios, with the results presented in the following table:

α	β	γ	X_1	X_2	X_3	U
0.1	0.6	0.3	2.2	13.2	6.6	89.63
0.2	0.5	0.3	4.4	11	6.6	78.57
0.3	0.4	0.3	6.6	8.8	6.6	74.05

Table 3: parameter of Lagrange Auxiliary Model

We here take the optimal combination $\alpha = 0.2, \beta = 0.5, \gamma = 0.3$ as an example, its 3D utility surface is shown in Figure 6.

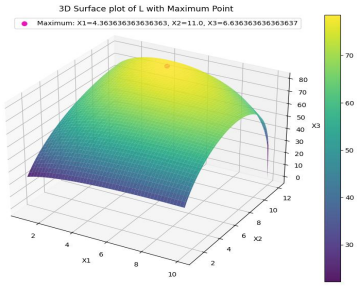


Figure 6: 3D Surface plot of L with maximum Point

4 Task 2: Player Value Evaluation and Roster Optimization Model

4.1 Notations

Symbol	Description
V_i	Comprehensive net value of player i measuring their total contribution to the team.
S_i	Synthetic competitive rating of the player i.
S^*	Rating benchmark for elite players(the 95th percentile)
S_{rep}	Rating benchmark for marginal rotation players at the replacement level (last 10%).
k	Win rate conversion coefficient mapping the competitive range to specific seasonal win increments.
θ_i	Market demand adjustment coefficient reflecting the shifting effect of the player i on the ticket demand curve.
β_{win}	the elasticity coefficient of revenue to the change in winning rate.
p_i	Probability of injury for the player i during the current season.
x_i	Binary decision variable indicating whether the player i is selected for the final roster.
B	Financial ceiling available for player salaries and signing costs.
K	Mandatory requirement for selection size as stipulated by league regulations.

Table 4: Notations used in 5.2, 5.3

4.2 Player Value Evaluation and Roster Optimization Model

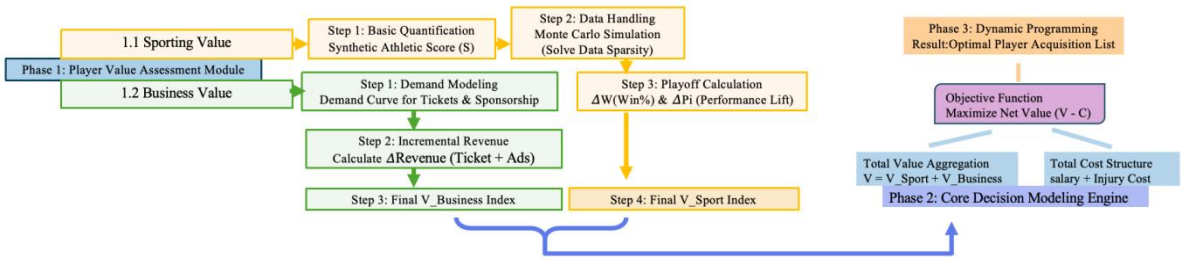


Figure 7: The Player Acquisition & Optimization Process

This model constructs a comprehensive evaluation system by quantifying the incremental revenue and incremental costs generated by a single player for the team, serving as the decision basis for roster configuration.

4.2.1 Decomposition of Individual Player Value

The net value V_i of a single player i is defined as the difference between the total incremental revenue and the total incremental costs they generate.

$$V_i = (V_i^{sport} + V_i^{business}) - Cost_i$$

4.2.1.1 Competitive incremental value V_i^{sport}

V_i^{sport} aims to quantify the marginal contribution of an individual player to the team's overall winning rate:

$$V_i^{sport} = \alpha \cdot \Delta W_i + \beta \cdot \Delta P_{playoff,i}$$

To accurately calculate these parameters, we first quantify the players by synthesizing the competitive score S.

Then we simulated the situations of ten thousand types of players through Monte Carlo simulation, and based on this simulation distribution, we defined the following key parameter logic to calculate ΔW_i.

Next, we defined the competitive difference factor $\text{diff}^* = S^* - S_{rep}$ to quantify the significant competitive performance gap between ordinary players and star players and win rate conversion coefficient calculated based on different winning rate targets $k = \frac{\text{win goals}}{\text{diff}^*}$, using to calibrate the conversion efficiency coefficients under different competitive states. Ultimately, the increase in the winning rate of a specific player can be calculated through Formula

$$\Delta W_i = k \cdot (S_i - S_{rep})$$

Then we define R_0 as the team's annual benchmark total revenue. α is derived from Formula

$$\alpha = R_0 \cdot \beta_{win} \cdot (1/N_{games})$$

Then we define baseline probability P_0, That is, the probability of a team making it to the playoffs under the initial number of wins W_0. And calculated benchmark probability P_1, by introducing ΔW and following the formula

$$playoff_{prob}(W) = \frac{1}{1 + \exp(-a \cdot (W - W_{\text{cut}}))}$$

Finally, ΔP is obtained, which is the difference between P_1 and P_0, reflecting the impact of changes in the number of wins on the probability of making the playoffs.

After obtaining all these values, we can finally measure the Competitive incremental value V_i^{sports}.

4.2.1.2 Commercial incremental value $V_i^{business}$

Commercial incremental value $V_i^{business}$ captures the non-competitive financial contributions that players bring to the team through their market appeal. We believe that a player's form on the field i will significantly regulate the demand curve of home games through their personal influence coefficient θ_i. The demand function Q for a single game is defined as follows:

$$Q = D(P, \theta_i) = a - bP + \theta_i$$

Here P is the average price of ticket. This demand expansion driven by "popularity" eventually translates into the team's actual income profits through two types of financial paths. Therefore, we define the commercial incremental value brought by :

$$V_i^{business} = P \cdot \theta_i + S_1 \cdot \theta_i$$

The box office increment $P \cdot \theta_i$ is derived by comparing the revenue differences when players are on the field and when they are absent ($\theta_i = 0$). The calculation formula is

$$\Delta R_i^{ticket} = P \cdot (D(P, \theta_i) - D(P, 0))$$

And the increment of commercial sponsorship $S_1\theta_1$ represents the marginal contribution of players to the total amount of team merchandise and sponsorship through social media interaction, jersey sales, and personal endorsements.

4.2.1.3 Incremental cost $Cost_i$

It mainly includes the players' salary expenses, transfer fees and signing bonuses, which constitute the negative subtraction term in the model.

4.2.2 Risk Assessment: Expected Injury Loss

This model's core logic involves converting potential negative events into anticipated financial discounts to adjust the final net value V_i of a player. To compensate for the limitations of static evaluation models when facing real-world operational risks, we further introduced the injury risk term $Risk_i$.

We define the expected injury loss for a specific player i as the product of the probability of injury p_i and the expected economic loss L_i resulting from such an injury:

$$Risk_i = p_i \cdot L_i$$

After incorporating this risk factor into the model, management's comprehensive evaluation of players no longer simply pursues the maximization of V_i^{sports} and $V_i^{business}$ but instead seeks the optimal equilibrium between risk and reward.

Here, we have also generated two visual images of the impact of injuries on the team through Monte Carlo simulation, which can more intuitively reflect this process:

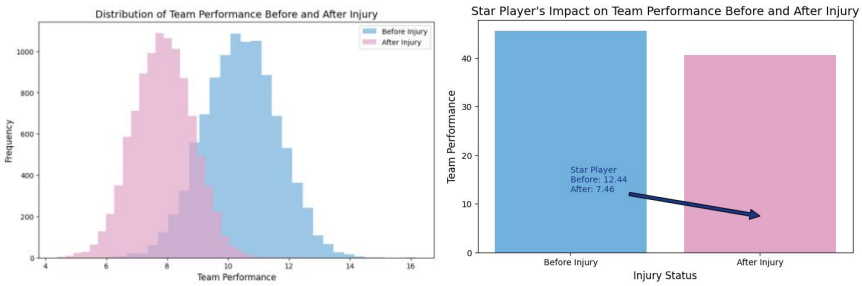


Figure 8: Team performance after injury

4.2.3 Decision Mechanism: Roster Selection Optimization

In this model, our goal to maximize the total net value of all the players on the team and ensure that the overall lineup achieves the maximum synergy in terms of competitive success and brand asset appreciation. Considering that in actual operation, the selection decision is rigidly constrained by the budget ceiling and the size of the lineup, we abstract the lineup selection as the following mathematical problem:

$$\max \sum_{i=1}^{N} y_i V_i$$

$$\text{s.t.} \begin{cases} \sum_{i=1}^{N} y_i Cost_i \leq B \\ \sum_{i=1}^{N} y_i \leq K \\ y_i \in \{0,1\} \end{cases}$$

4.3 The Solution of Model 2

This model aims to establish an evaluation criterion for player value and a dynamic selection mechanism. Based on the evaluation criteria, our budget, and the limit on the number of selected players, the selection of players is carried out. Due to the limited number of players in the candidate pool, we ultimately conducted a simulation experiment on the model using the data of twenty players in the data. Here, we set the total budget at 7.1M. The upper limit of the number of selected people is 5. The following three people will be ultimately selected:

Player	Contract Type	Years of Service	Salary 2025	All Star Selections	Championship	V Total
Brittney Griner	Max	11	214466	9	2	9.048364
A'ja Wilson	Below Max	7	200000	6	3	6.971563
Jewell Loyd	Max	10	245508	5	1	6.044457

Table 5: Information of the selected players in the example

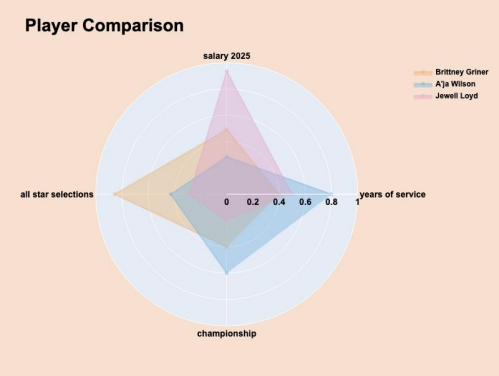


Figure 9: player Comparison Score

From this, we can see that when selecting players, indicators related to their performance on the field are particularly important for the overall score. And when the team's budget is limited, choosing players who perform outstandingly but are under the "maximum contract" is also a feasible strategy.

5 Task 3 : Player Net Value Evaluation under League Expansion

5.1 Model Overview & Objectives

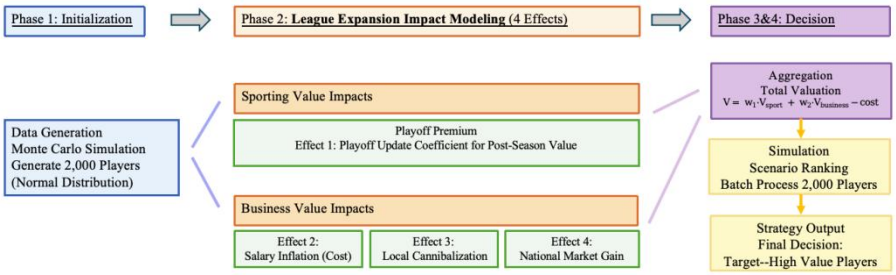


Figure 10: The Expansion Impact & Scenario Simulation Model

To cope with the possible expansion of the WNBA league, this model aims to evaluate the impact of different expansion scenarios on the Average Net Value of players. By simulating the joining of new teams and market expansion, we assist team management in formulating the best operational strategies in complex environments such as resource diversion and salary inflation.

5.2 Notations

Symbol	Description
W_i	Win Contribution
$P_{playoff}$	Playoff Probability
κ	National Gain Coefficient
S_i	Endorsement Factor
P_{ko}	Exposure Conversion
η	The cost of talent competition has risen due to the expansion of the alliance
β	The intensity of the team's competitive goals
λ	The speed at which the expansion of the military diluts the original local resources of the team
base salary	Benchmark variable on the cost side of players
d_i	The spatial relationship between new and existing football markets

Table 6: Notations used in 6.3,6.4

5.3 Stochastic Player Performance Generation

Due to the uncertainty of players' performance, we first simulate the basic data of the player database through a random process. To ensure the authenticity of the simulation, we assume that a player's "win contribution" and "playoff potential" are not independent of each other but have a certain positive correlation.

We utilize the multivariate normal distribution to generate the associated competitive indicators and construct the covariance matrix Σ to maintain the correlation (We set $\rho = 0.3$). To ensure that the generated data conforms to the actual sports logic, we truncated the

originally generated variables.

5.4 Mathematical Formulation of Market Effects

5.4.1 Local Market Leakage

Based on the principle of the gravitational model, the addition of a new team will generate market appeal, and its intensity is exponentially negatively correlated with spatial distance. As the distance increases, the team's influence on the remote market declines exponentially. Therefore, we define the split effect function as follows:

$$leakage_effect = \lambda \times \exp\left(\frac{-d_i}{h}\right)$$

The distance decay scale h defines the influence boundary of geographical competition,When a new team enters a mature market, d_i is relatively small, and the core market will face a fierce zero-sum game. As d_i increases, the negative impact will rapidly diminish.

5.4.2 National Market Gain

Correspondingly, the expansion of the league will significantly enhance the marginal value of the league, and the addition of new teams will increase the overall exposure of the league. We define the calculation formula for the national gain factor as:

$$national_gain = (1 + \kappa \times MP_new_team)$$

Here, MP_new_team represents the economic volume and potential of the new market, while κ measures the efficiency of converting this increment into the commercial value of the existing team.

5.4.3 Net Value Calculation

To assess the strategic asset value of players in the context of league expansion, we integrate the competitive output, commercial dividends and financial costs derived in the previous text. Based on the value distribution logic determined in 4.1.2, the comprehensive net worth of individual player i is redefined as NV_i:

$$NV_i = \underbrace{\alpha \cdot win_contrib_i + \beta \cdot playoff_prob_i}_{V_i^{sport}} + \underbrace{business_value_i}_{V_i^{business}} - \underbrace{Salary_i}_{Cost_i}$$

Among them, the dynamic assessment of competitive value V_i^{sport}: We still retain the contribution to wins and the probability of making the playoffs as the core of competition.

Environmental correction of business value $V_i^{business}$: This is the core innovation point of the model. It combines the players' popularity S_t, the exposure conversion coefficientP_{ko}, and the market correction factor driven by the expansionλ. It quantifies whether the national market gain can offset the resource diversion in the local market.

5.5 The Solution of Model 3

This model constructs a scenario simulation framework, which assesses the average net value of players under different macro business assumptions to determine which type of strategy combination can maximize the team's comprehensive return in the long term. In this framework, the team, by adjusting five core decision-making parameters, jointly deduces the optimal strategic choice under multiple resource constraints from the dimensions of cost pressure, competitive intensity, resource diversion, financial flexibility, and geographical expansion boundaries. Here, we simulated the basic data of over 2,000 players that conform to the normal distribution through Monte Carlo simulation and adjusted the parameter assignment based on the decision-making situation. Ultimately, under different decision-making scenarios, the average comprehensive scores of over 2,000 players were compared.

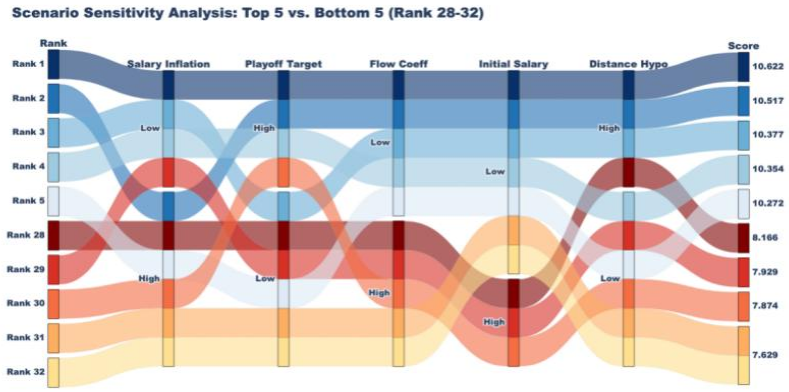


Figure 11: Scenario Seneitivity Analysis: Top 5 vs. Bottom 5

This is the result of our scenario simulation. Here we find that a high diversion coefficient and a high playoff target play a positive role in this type of strategy. This may imply that recruiting players who perform better in the game through highly concentrated resource investment is conducive to achieving higher event goals, and higher goal setting can generate more game and commercial income. This is also beneficial to the average value increase of players, forming a virtuous cycle.

As for salary, we have found that a low initial salary may bring about favorable results. This might indicate that low initial salary corresponds to more welfare benefits, and these welfare benefits are more attractive to players. But he situation of salary inflation varies greatly, and it is impossible to draw very significant and convincing conclusions from it.

6 Task 4: Zone-Based Dynamic Pricing Model with LTV Integration

6.1 The "Profit-Acquisition" Trade-off

The traditional sports pricing model usually only pursues the maximization of ticket revenue, which often leads to poor attendance at unpopular events, thus losing the opportunity for secondary consumption and the cultivation of long-term fans. For LV Aces, we have proposed a hierarchical dynamic pricing model. The core innovation lies in dividing the venue into three strategic functional areas and setting different objective function weights:

- High/VIP Zone: The profit-harvesting area.
- Mid/Standard Zone: Balance buffer zone.
- Low/Value Zone: Customer acquisition and conversion area.

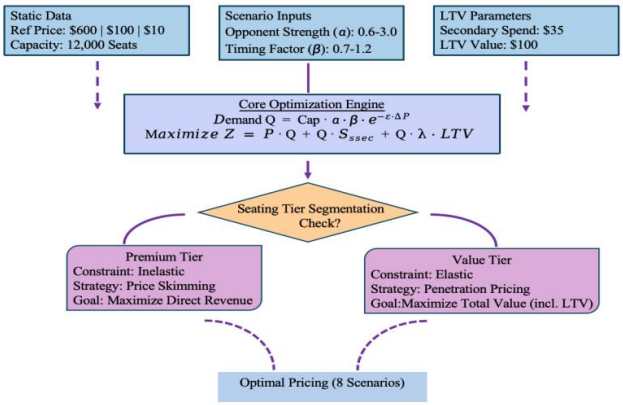


Figure 12: The Strategic Pricing & Profitability Model

6.2 Notations

Symbol	Description	Unit/Range
z	Zone index (High, Mid, Low)	$z \in \{High, Mid, Low\}$
s	Scenario index (Opponent/Timing)	$\alpha \in [0.6, 3.0], \beta \in [0.7, 1.2]$
$P_{z,s}$	Decision Variable: Ticket price for zone z in scenario s	USD ($)
$P_{ref,z}$	Reference price based on 2025 season data	High: $600, Mid: $100, Low: $10
$\epsilon_{z,s}$	Price elasticity of demand	High ≈ 0.6, Low ≈ 2.5
S_{sec}	Average secondary spending (F&B/Merchandise)	$35 per attendee
LTV	Expected potential Lifetime Value	$100 per new fan converted
λ_z	Acquisition weight (applied only to Low Zone)	$\lambda_{Low} \approx 1, \lambda_{High} = 0$

Table 7: Notations used in 7.3

6.3 Model construction

6.3.1 The revised demand function

We adopt the log-linear model to simulate the demand and introduce the truncation function to reflect the physical capacity limit Cap_z of Michelob ULTRA Arena.

$$Q_{potential}(P_{z,s}) = Cap_z \cdot \underbrace{\alpha_s \cdot \beta_s}_{\text{External Factors}} \cdot \exp\left(-\epsilon_{z,s} \cdot \frac{P_{z,s} - P_{ref,z}}{P_{ref,z}}\right)$$

$$Q_{sold}(P_{z,s}) = \min(Q_{potential}(P_{z,s}), \quad Cap_z)$$

Logically, demand is driven by external factors (the popularity of rivals α, time coefficient β) and decays exponentially with the extent to which the price deviates from the reference price.

6.3.2 Objective function: Maximizing total utility

The innovation point of this model lies in maximizing the total utility(Z), which consists of three parts: direct ticket revenue, secondary consumption revenue and long-term customer acquisition value.

$$\text{Maximize } Z = \sum_z \left[\underbrace{(P_z \cdot Q_{sold})}_{\text{Immediate Revenue}} + \underbrace{(Q_{sold} \cdot S_{sec})}_{\text{Secondary Revenue}} + \underbrace{(Q_{sold} \cdot \lambda_z \cdot LTV)}_{\text{Future Asset Value}} \right]$$

To ensure the credibility of the model, LTV ($100) is calculated through the following probability weighting formula:

$$LTV_{Low} = \underbrace{(P_{ret} \cdot N_g \cdot M_t)}_{\text{1. Future Retention}} + \underbrace{(P_{merch} \cdot M_m)}_{\text{2. Merch Upsell}} + \underbrace{(P_{season} \cdot M_s)}_{\text{3. Season Ticket Conv.}} + \underbrace{V_{data}}_{\text{4. Data Value}}$$

Among them, Future Retention indicates that based on a repurchase rate of 20%, the average number of games watched is 3. Merch Upsell stated that based on a 15% on-site retail conversion rate. Season Ticket Conv. Indicates a 5% sales funnel conversion rate. The Data Value is set at $15.5 with reference to the industry customer acquisition cost (CPL).

6.4 The Solution of Model 4

To verify the multi-scenario adaptability of the model, we constructed eight typical scenarios and obtained the optimal pricing strategies for each scenario.

Scenario	α	β	High (VIP)	Mid (Std)	Low (Value)	Baseline Utility	Z (Max)	Gain%
1.The Clark Effect	2.8	1.2	$2,926	$293	$19	2,475,000	6,882,484	178
2.Champ Rivalry	2	1.1	$1,818	$202	$15	2,475,000	4,769,932	93

3.Season Opener	2.2	1	$1,995	$214	$16	2,475,000	5,097,732	106
4.Weekend Family	1.2	1.1	$1,020	$136	$12	2,475,000	3,263,496	32
5.Winning Streak	1.4	0.9	$1,079	$138	$12	2,475,000	3,346,422	35
6.Weekday Grinder	1.1	0.7	$740	$94	$10	2,286,900	2,387,358	4
7.Fan Appreciation	0.8	1	$636	$99	$10	2,376,000	2,420,511	2
8.Tuesday Slump	0.6	0.7	$518	$69	$8	1,247,400	1,849,334	48

Table 8: Simulation results for 8 situation

In this table, “baseline utility” represents the revenue that can be obtained if the initial baseline fare is charged under different scenario simulation conditions.

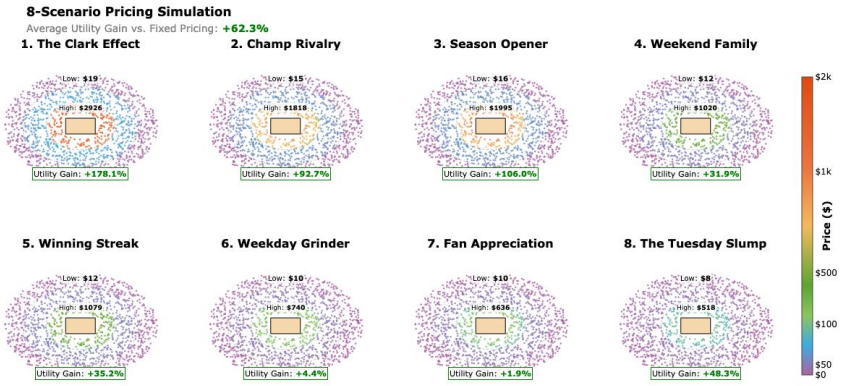


Figure 13: 8-Scenario Pricing Simulation

Overall whether in favorable or adverse circumstances, the model demonstrates the ability to automatically recalibrate ticketing strategies based on real-time demands. This is most evident in the comparison of extreme cases: in Scenario 1 (championship contenders), the model aggressively raised the price point to maximize revenue by taking advantage of the passionate interest of fans; In Scenario 8 (the reconstruction phase), the model significantly reduced the price, prioritizing the stability of the occupancy rate to protect the long-term fan base. This adaptive behavior verifies that regardless of how the team's performance on the field fluctuates, the model can effectively guarantee financial health. Meanwhile, we found that this pricing model can increase the Utility of the original time by up to 62.3%, which also proves the validity of the model.

7 Model Evaluation and Further Discussion

7.1 Strengths

- **Dynamic decision optimization:** Integrates dynamic trade-offs and Lagrangian methods to capture the evolving interaction between performance and finances under multiple scenarios.
- **Data-driven valuation:** Combines on-court performance and off-court indicators (e.g., All-Star selections, social media presence) to reflect player market value.
- **Granular resource allocation:** Provides refined allocation strategies across player tiers and seasonal objectives under budget constraints.
- **Scalable and adaptable:** Incorporates expansion and injury risks, enabling flexible strategy adjustments across future scenarios.

7.2 Weaknesses

- **Limits from simplified assumptions:** Constant growth assumptions limit the model's ability to capture uncertainty and unexpected shocks.
- **Individual-focused valuation bias:** Emphasis on star players and subjective scoring underrepresents team chemistry and roster structure.
- **Limited empirical validation:** Further testing with real game and financial data is needed to improve scalability and practical use.

7.3 Further Discussion

While the current model provides a solid foundation for optimizing team performance and revenue outcomes, there are still clear directions for improvement.

First, future work could incorporate time-series analysis and machine learning–based forecasting to uncover deeper patterns in historical data. This would allow the model to move beyond static evaluation and toward trend prediction, making it better at capturing performance volatility and changes in the market environment over time.

Second, the model could be strengthened through tighter validation against real-world data by continuously calibrating parameters using actual season outcomes. In particular, bringing in external factors such as social media sentiment, media coverage, and fan engagement as moderating variables would expand the evaluation framework and improve the model's applicability in more complex, real-world settings.



February 2, 2026
To the Owner and General Manager Las Vegas Aces Las Vegas, NV
RE: Strategic Recommendations for Roster and Business Decisions in the Upcoming Season

Dear Ownership and General Management,

It is our honor to submit this strategic proposal for the future development of the Las Vegas Aces. This letter presents an integrated business management framework for the upcoming season that aligns roster construction with financial operations, with the objective of simultaneously supporting the Aces' on-court success and long-term financial health.

I. At the core of our strategic approach, we seek to enhance the franchise's long term value by optimizing resource allocation:

Marketing investment should take precedence over pure salary escalation. Rather than engaging in increasingly aggressive salary competition under a constrained salary cap, the team should reallocate marginal resources toward marketing and brand development. Strengthening brand equity generates nonlinear returns by amplifying both player attraction and revenue potential beyond direct payroll effects.

Lock in the optimal configuration of a high market flow coefficient, a high playoff target, and low initial salary commitments. Simulation results indicate that the combination of high market flow and elevated playoff aspirations plays a decisive role in attracting top-tier talent, as higher competitive targets significantly enhance player appeal while magnifying ticketing and commercial revenue effects. In contrast, lower initial salary offers create greater flexibility for benefits-based compensation, forming a more adhesive recruitment mechanism during league expansion periods. Among the modeled factors, salary inflation exhibits relatively limited marginal impact, while geographic distance decay remains an exogenous constraint beyond organizational control.

Reorient ticket pricing toward maximizing fan lifetime value. Pricing should shift away from single-game revenue maximization toward a structure that acquires new fans through low-priced outer-bowl seating and monetizes them through in-arena secondary spending and long-term repeat engagement. Operationally, this requires abandoning linear venue pricing in favor of asymmetric, zone-based strategies, with premium pricing in core seating areas driving revenue and ultra-low pricing in peripheral zones driving attendance. Moreover, pricing must remain highly context-sensitive: aggressively capturing demand surges during championship windows and marquee matchups, while protecting the fan base through high value pricing during rebuilding phases or low-attention periods. In this framework, ticket pricing becomes a strategic lever for strengthening brand loyalty and sustaining long-term franchise value.



II. While pursuing a higher performance ceiling, the organization must simultaneously establish a resilient floor through disciplined trade-offs and risk management.

Prioritize franchise valuation over short-term net profit and adopt a long-term operating mindset. Although elevated payrolls and fixed costs may constrain annual net profits, franchise valuation can grow at a substantially faster pace. Management should tolerate short-term financial volatility and focus instead on expanding player market value and the fan base, which are the primary drivers of long-term franchise appreciation.

Adopt a "max-contract core plus high-value sub-max players" roster composition. When operating under budget constraints, targeting high-performing players whose salary demands fall below the maximum contract level represents a financially efficient strategy. Roster decisions should ensure that lower-salary players can deliver on-court contributions that partially substitute for higher-paid stars, thereby preserving roster depth without breaching financial limits.

Systematically embed risk-hedging mechanisms into roster construction, prioritizing tactical depth over reliance on a single star. Simulation results indicate that injuries to marquee players generate significant and nonlinear negative impacts on team performance. To mitigate this risk, the team should prioritize acquiring tactically versatile role players and quantify expected injury losses through calibrated impact coefficients, while reserving sufficient replacement-level capacity within the roster. Accordingly, the model favors protecting core players with higher durability and availability, reinforcing both competitive stability and long-term sustainability.

In the business logic of professional sports, competitive success and financial health are not opposing trade-offs, but mutually reinforcing elements of a dynamic system. When managed in coordination, on-court excellence can be systematically converted into long-term economic value, while a sound financial foundation, in turn, sustains competitive performance. This virtuous cycle positions the Las Vegas Aces to build a competitive advantage in the coming season that is both durable and difficult to reverse.

We believe this strategy will enable the Las Vegas Aces to remain a long-term contender both on the court and in the marketplace. The organization now faces a unique window to define the future trajectory of this franchise, and we would welcome the opportunity to discuss the underlying data and simulation insights in greater detail at your convenience.

Respectfully,
Strategic Operations & Analytics Department

9 References

[1] Basketball Reference. (n.d.). WNBA stats, history, scores, & records. Retrieved February 1, 2026, from https://www.basketball-reference.com/wnba/

[2] Women's National Basketball Association. (n.d.). WNBA official website. Retrieved February 1, 2026, from https://www.wnba.com/

[3] The Associated Press. (2024, May 7). WNBA to begin full-time charter flights this season, commissioner says. CBC Sports. https://www.cbc.ca/sports/basketball/wnba-full-time-charter-flights-1.7197074

[4] Sportico. (2025). Sports valuations. The Business of Sports. Retrieved February 2, 2026, from https://www.sportico.com/valuations/

[5] WordStream. (2024). Google Ads Benchmarks for Your Industry [Updated].. Retrieved from https://www.wordstream.com/blog/ws/2016/02/29/google-adwords-industry-benchmarks

[6] Green, A., & Reeves, R. (2025). Who are the highest-paid coaches in the WNBA? Front Office Sports. https://frontofficesports.com/highest-paid-wnba-coaches/

[7] Snel, A. (2024, July 29). Aces increase season tickets for 2025; First increase since owner Mark Davis bought WNBA team from MGM Resorts Int'l in 2021. LVSportsBiz. https://lvsportsbiz.com/2024/07/29/aces-increase-season-tickets-for-2025-first-increase-since-owner-mark-davis-bought-wnba-team-from-mgm-resorts-intl-in-2021/

[8] Dopson, E. (2024). Retail Conversion Rates: Average Benchmarks by Industry. The Retail Blog. Retrieved from https://www.shopify.com/retail/retail-conversion-rate

[9] Koebert, J. (2026, January 13). The spending habits of NBA fans on their favorite teams and players [Survey]. FinanceBuzz. https://financebuzz.com/nba-fan-spending

[10] Munson, B. (2025, November 09). WNBA Revenue-Sharing CBA. https://www.spotrac.com/news/_/id/3067/wnba-revenue-sharing-cba-debate

[11] Muret, D. (2023). Data and analytics continue to drive ticket sales and retention strategies for teams. Sports Business Journal. Retrieved from https://www.sportsbusinessjournal.com

[12] Ruler Analytics. (2023). Average Conversion Rate by Industry: The Ultimate Guide. Retrieved from https://www.ruleranalytics.com/blog/reporting/average-conversion-rate-by-industry/

Report on Use of AI

1. Gemini 3

Query 1: How many teams are in WNBA now and summarize the 2026 WNBA expansion draft rules.

As of 2026, the WNBA consists of **15 teams**, following the addition of the **Golden State Valkyries** (who debuted in 2025) and the **Toronto Tempo** and **Portland Fire** (debuting in 2026).

The rules for the 2026 expansion draft (designed for the Toronto and Portland franchises) are largely modeled after the 2024 expansion draft, though some specifics remain subject to the finalized Collective Bargaining Agreement (CBA). Below is a summary of the draft rules and procedures:

2026 Expansion Draft Summary

* **Expansion Teams**: The **Toronto Tempo** and **Portland Fire** are the two teams participating as they prepare for their inaugural 2026 season.
* **Protected Player Lists**: Each of the 13 existing WNBA teams must submit a "Roster List" of all players to whom they hold rights. From this list, teams are permitted to protect up to **six players** who cannot be selected in the expansion draft.
* **Draft Pool**: Any player not designated as "protected" is available for selection. This includes active players, suspended players, and those on retired or reserved lists.
* **Selection Limits**:
* Expansion teams can typically select **one player** from each existing team.
* Existing teams cannot lose more than one player total during the expansion process.
* **Unrestricted Free Agents (UFAs)**: Expansion teams may select at most **one player** who is set to become an unrestricted free agent. However, that player is only eligible if they have not previously played under a "Core Player" contract for two or more seasons.
* **Trades and Transactions**: Expansion teams are allowed to negotiate trades before the draft, such as agreeing to select (or not select) a specific player in exchange for assets like future draft picks.
* **Draft Order and Allocation**: To determine the order of picks between the two new teams, the WNBA may use a coin flip to decide which team receives the first pick in the expansion draft versus higher positioning in the subsequent rookie draft.